

DAY 00 / MASTER INDEX

Foundation Collection

Topic Index

A guided map of the complete 7-day collection: networking foundations, routing, automation, cloud platforms, safe operations, technical leadership, and an industry capstone.

7	493	29,056
TOPIC NOTES	PDF PAGES	SOURCE LINES

RECOMMENDED PATH

- 1

Build the networking foundation

Days 1-2: networking foundations, routing, data-centre networking, VXLAN/EVPN, WAN and SD-WAN.
- 2

Build and operate the platform

Days 3-5: network automation, Kubernetes and cloud platforms, SRE, observability, security and resilience.
- 3

Lead and apply the complete system

Days 6-7: architecture, strategy, governance, technical leadership, and the industry capstone.

HOW TO USE THIS INDEX

Choose a topic below, open the matching Markdown source for editing, or use the companion PDF in Foundation/pdfs for reading and review.

MODULE 01 / DAYS 01-02

Networking foundations and modern routing

Start with the packet-delivery model, then extend it into enterprise routing, data-centre fabrics, overlays, and WAN connectivity.

DAY

01

Networking Foundations for Software and Platform Engineers

A network exists so that one process running on one machine can exchange data with another process, potentially running on another machine thousands of kilometres away.

KEY COVERAGE hosts and interfaces; MAC and IP addresses; ports and sockets; OSI and TCP/IP; Ethernet; routing; TCP/UDP; DNS; firewalls; TLS; load balancers

SOURCE [Day-1-Networking-Foundations-for-Software-and-Platform-Engineers.md](#) | **PDF** 84 pages

DAY

02

Routing, Data-Centre Networking, VXLAN/EVPN, WAN and SD-WAN

Modern networks separate physical connectivity from logical connectivity, and they separate learning decisions from packet forwarding.

KEY COVERAGE routing tables and longest-prefix match; OSPF and BGP; ECMP; leaf-spine; underlays and overlays; VRF/VNI; VXLAN; EVPN; MPLS; WAN and SD-WAN

SOURCE

[Day-2-Routing-Data-Centre-Networking-VXLAN-EVPN-WAN-and-SD-WAN.md](#) | **PDF** 60 pages

NETWORK MEMORY LINE

Applications communicate through names and ports; routing finds paths between networks; overlays create logical networks above resilient IP connectivity.

MODULE 02 / DAYS 03-05

Automation, cloud platforms, and safe operations

Build the infrastructure-control platform, run it on durable distributed foundations, and operate it with reliability, security, observability, and recovery controls.

DAY

03

Network Automation, APIs, Orchestration and Infrastructure as Code

A modern network-automation platform is much closer to a distributed infrastructure-control platform than a script that runs commands.

KEY COVERAGE automation evolution; inventory, IPAM and CMDB; source of truth; idempotency; APIs and YANG; IaC and GitOps; queues; workflow engines; safe rollout; closed-loop automation

SOURCE Day-3-Network-Automation-APIs-Orchestration-and-Infrastructure-as-Code.md | **PDF** 72 pages

DAY

04

Kubernetes, Cloud, Distributed Platforms, CI/CD and Data

A production network-automation system is much more than an API that sends commands to routers.

KEY COVERAGE containers and Kubernetes; declarative workloads; networking and storage; distributed systems; databases and caching; CI/CD and IaC; cloud and hybrid platforms; telemetry and data

SOURCE Day-4-Kubernetes-Cloud-Distributed-Platforms-CI-CD-and-Data.md | **PDF** 66 pages

DAY

05

SRE, Observability, Security, Risk, Controls and Resilience

A production platform is good because it behaves predictably when things fail, tells us what is happening, limits damage, protects critical assets, and can recover safely.

KEY COVERAGE reliability vocabulary; failure domains; SLI/SLO/SLA; golden signals; logs, metrics and traces; alerting; failure testing; identity and access; threat modelling; audit; disaster recovery

SOURCE Day-5-SRE-Observability-Security-Risk-Controls-and-Resilience.md | **PDF** 68 pages

PLATFORM MEMORY LINE

Declarative desired state + durable distributed state + replaceable compute + controlled automation + continuous reconciliation.

MODULE 03 / DAYS 06-07

Technical leadership and industry application

Connect architecture, strategy, governance, people, economics, and risk, then apply the complete model to a global network-automation capstone.

DAY

06

Lead/VP Engineering: Architecture, Strategy, Governance and Technical Leadership

At Lead/Staff/Principal/VP level, engineering increasingly means creating an environment in which many teams can repeatedly produce correct, secure, reliable and economically sensible systems.

KEY COVERAGE problem framing; requirements; strategy; architecture trade-offs; platform engineering; technical debt; governance; risk; stakeholders; quality; economics; capacity; organisation and roadmap

SOURCE Day-6-Lead-VP-Engineering-Architecture-Strategy-Governance-and-Technical-Leadership.md | **PDF** 60 pages

DAY

07

Building and Operating a Global Network Automation Platform

A fictional industry case study designed to resemble the engineering environment and responsibilities of a lead-level engineer in a large regulated global bank.

KEY COVERAGE business problem; requirements and architecture; workflows and queues; safe change; security and controls; Kubernetes deployment; observability; production incident; disaster recovery; scale and governance

SOURCE Day-7-Industry-Capstone.md | **PDF** 83 pages

LEADERSHIP MEMORY LINE

Connect technology, architecture, reliability, security, risk, economics, people and strategy into one sustainable engineering system.

QUICK LOCATOR

Find a topic fast

Use this cross-reference when you know the concept you need but not the day that covers it.

Need to find...	Start with	Best note
Network layers, IP, ports, DNS	Networking foundations	Day 01
OSPF, BGP, EVPN, VXLAN, WAN, SD-WAN	Routing and overlays	Day 02
APIs, source of truth, IaC, safe rollout	Network automation	Day 03
Kubernetes, cloud, distributed data, CI/CD	Platform foundations	Day 04
SRE, observability, security, resilience	Safe production operations	Day 05
Architecture, strategy, governance, leadership	Lead/VP engineering	Day 06
Global network automation platform	Industry capstone	Day 07

COLLECTION SUMMARY

Source format	7 Markdown files in Foundation/
Reading format	7 companion PDFs in Foundation/pdfs/
Total coverage	29,056 source lines and 493 PDF pages
Index generated	8 August 2026